Atomic Drift Tuner v0.8.0-beta.1 — How to Use

Atomic Drift Tuner is a Windows tuning assistant for Assetto Corsa drifting. It combines wheelbase and steering-wheel information, drift-car and drift-pack context, driver intent, live Assetto Corsa telemetry, MOZA/AZOM recommendations, Assetto Corsa FFB recommendations, per-car setup recommendations, per-car Desired Behavior targets, and an optional local-network iPhone/browser remote.

The Windows app remains the main controller. Atomic Remote on your phone is a companion interface and does not talk directly to your wheelbase, SimHub, AZOM, or Assetto Corsa.

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1. What you need

For the full feature set:

- Windows 10 or Windows 11
- Assetto Corsa
- SimHub
- AZOM
- supported MOZA hardware
- Atomic Drift Tuner
- Atomic SimHub Bridge v0.7.2 for live AZOM read/write features

You can still use many Atomic features without the bridge, including generated recommendations, AC car scanning, AC setup tuning, Desired Behavior, and telemetry-based analysis.

2. First launch

Open Atomic Drift Tuner.

On first run, use **Setup & Paths** to confirm:

- Assetto Corsa installation path
- Assetto Corsa setup folder
- SimHub path
- any other machine-specific paths shown by Atomic

If Atomic detects your Assetto Corsa install automatically, verify the detected path before continuing.

3. Select your hardware

On the main Atomic window, choose your wheelbase and steering wheel.

Built-in wheelbase profiles include MOZA R3, R5, R9, R12, R16, R21, R25 Ultra, and Custom DD.

Built-in wheel/rim options include CS Pro, KS Pro, CS V2P, RS V2, KS, ES / ES Lite / ESX, Vision GS, GS V2P, TSW, and Custom / aftermarket.

Wheel size, mass, and inertia influence Atomic's generated recommendations.

4. Let Atomic detect your Assetto Corsa car

Atomic v0.8 can automatically scan your installed cars and follow the car currently loaded in Assetto Corsa.

Recommended options:

- **Auto-scan installed cars when Atomic starts / AC path changes**
- **Auto-select active AC car + inferred drift pack**

Leave both enabled unless you specifically want manual control.

When Assetto Corsa is on track, Atomic reads the current AC session identity and matches the loaded car to your installed `content\cars\<car-folder>`.

Atomic prefers exact matching rather than fuzzy guesses.

When a match is found, Atomic can automatically update:

- current car
- drift pack
- Windows tuning context
- iPhone / browser remote context

If Atomic cannot confidently identify the pack, it uses **Custom / Other Pack** instead of guessing.

5. Drift-pack detection

Atomic can infer known drift packs from installed-car metadata and folder information.

Known pack categories include:

- VDC
- Gravy Garage
- Team SWARM
- ADL
- WDT / WDTS
- Deathwish Garage
- Custom / Other

You can always override the detected pack manually if needed.

6. Choose your Drift Target

The **Drift Target** tells Atomic what type of behavior you want from the current session.

Available targets include:

- Training
- Realistic
- Fast Self-Steer
- Tandem
- Competition

The Drift Target has higher priority than the per-car Desired Behavior sliders.

If the Drift Target and Desired Behavior pull a setup parameter in opposite directions, Atomic blends the two rather than blindly stacking conflicting changes.

7. Generate a tune

After selecting or auto-detecting your wheelbase, steering wheel, drift pack, car, and Drift Target, click **Generate Tune**.

Atomic can generate AZOM / wheelbase recommendations such as:

- Game FFB Strength
- Base Torque Output
- Wheel Rotation
- Maximum Wheel Speed
- Interpolation
- Wheel Damper
- Wheel Friction
- Natural Inertia
- High-Speed Damping
- High-Speed Trigger

It can also generate Assetto Corsa FFB recommendations such as:

- Gain
- Filter
- Minimum Force
- Kerb
- Road
- Slip
- ABS

Atomic also produces summary scores such as Self-Steer, Stability, and Detail, and may show estimated peak wheel torque and tuning notes.

Generating a tune does **not** automatically apply settings to the wheelbase.

8. Review Full AZOM Settings

Open **Full AZOM Settings** to inspect the complete generated recommendation.

When SimHub, AZOM, the MOZA base, and Atomic SimHub Bridge v0.7.2 are available, Atomic can read live AZOM values.

Use **Read Live AZOM** to compare current live values with Atomic's recommended values.

Atomic's guarded AZOM write system includes:

- only supported settings are written
- only changed settings are written
- one write batch at a time
- duplicate-target suppression
- direct-write spacing
- exact AZOM commit path
- live readback verification
- stop on first unverified write
- pre-apply snapshot

- Revert support

If anything behaves unexpectedly, stop driving and revert.

9. Revert AZOM changes

Use **Revert Last Apply** to restore the settings changed by the most recent Atomic apply.

Atomic does not blindly restore unrelated settings.

10. Use the AC Car Setup Tuner

Open **AC Car Setup Tuner** when you want Atomic to help tune the in-game car setup.

Atomic can:

- inspect installed car data
- inspect available setup ranges
- read a baseline AC setup
- generate recommended setup changes
- explain the reason for each change
- show current / recommended / delta
- export a new setup

Atomic writes new files using names such as `Atomic_*.ini` and does not overwrite your original baseline setup.

11. Desired Car Behavior

Desired Behavior lets you tell Atomic how you want a specific car to feel.

Behavior controls include:

- Front-End Bite
- Rear Grip
- Self-Steer Speed
- Transition Speed
- Angle Stability
- Throttle Steering
- Initiation Sharpness

Presets include:

- Neutral
- Stable & Forgiving
- Fast Tandem
- Fast + Stable
- Aggressive Rotation

After changing Desired Behavior, save it for the current car. Atomic will use that profile when generating AC setup recommendations for that car in the future.

12. How Atomic blends conflicting behavior goals

Some requested behaviors push the same setup parameter in opposite directions.

Atomic can:

- reduce overlapping effects
- apply diminishing returns
- identify compromises
- preserve the higher-priority Drift Target
- explain the compromise in the setup recommendation

Possible blend states include:

- Single goal
- Aligned / damped
- Compromise
- Intent compromise
- Behavior + intent compromise
- Balanced out

13. Live Assetto Corsa telemetry

Open **Telemetry Recorder** while Assetto Corsa is running and you are in an on-track session.

Click **Connect**.

Atomic can display values such as:

- speed
- slip angle
- steering angle
- FFB output
- drift state

Atomic's current drift heuristic considers the car to be drifting when speed is at least 20 km/h and absolute body slip angle is at least 10 degrees.

Telemetry can be recorded for later analysis.

14. Tuning Assistant

Use **Tuning Assistant** after collecting telemetry and driving feedback.

Atomic can combine:

- recorded telemetry
- current tune
- hardware
- wheel
- pack
- car
- your feedback

Feedback categories include:

- self-steer
- FFB strength
- steering weight
- detail / noise
- oscillation

Atomic stores calibration separately for the exact hardware + wheel + pack + car combination.

15. Atomic Remote on iPhone or another browser

Open **Remote / iPhone Test** on the Windows app.

To connect:

1. Leave **Remote AZOM Writes** OFF at first.
2. Click **Start Remote**.
3. If Windows Firewall asks, allow Atomic on your private network.
4. Make sure the phone and PC are on the same local network.
5. Open the LAN address shown by Atomic in Safari or another browser.
6. Enter the six-digit pairing code shown on the Windows app.

Atomic Remote is intended for a trusted private LAN. Do **not** port-forward the Atomic Remote port or expose it directly to the public Internet.

16. Remote Dashboard

The phone Dashboard can show:

- current Atomic version
- current wheelbase
- steering wheel
- detected drift pack
- active car
- Drift Target
- live AC speed
- live slip angle
- live steering angle
- live FFB output

- drift detection
- telemetry connection state
- generated tune scores

Because Atomic uses a shared telemetry service, the Windows app and phone consume the same live AC telemetry stream.

17. Remote Tune tab

The **Tune** tab lets you:

- change the Windows Drift Target
- request Generate Tune
- review the generated recommendations

Changing the Drift Target from the phone updates the authoritative Windows selector.

Pressing **Generate Tune on Windows** runs the normal Atomic tuning engine on the PC. It does **not** automatically apply anything to AZOM.

The phone can then display:

- AZOM / MOZA recommendations
- AC FFB recommendations
- self-steer score
- stability score
- detail score
- estimated peak wheel torque
- notes

18. Remote Behavior tab

The **Behavior** tab lets you edit the Desired Behavior profile for the currently selected or auto-detected car.

You can move the behavior sliders, load a preset, and save the behavior profile for that car.

The phone writes the profile back to the normal Windows Atomic behavior-profile store. It does not directly change the wheelbase.

19. Remote AZOM tab

The **AZOM** tab can display selected live AZOM values.

Remote hardware writes are intentionally limited.

Remote AZOM writes are:

- OFF every time the remote server starts

- enabled only from the Windows PC
- limited to a conservative allow-list
- passed back through the existing Windows Atomic write safeguards

The phone never talks directly to AZOM or the MOZA base.

20. Safe remote AZOM testing

When testing remote write control:

1. Start SimHub.
2. Start AZOM.
3. Confirm the MOZA base is connected.
4. Confirm Atomic can read live AZOM settings.
5. Start Atomic Remote.
6. Pair the phone.
7. Enable remote writes on the Windows PC only when ready.
8. Change one conservative setting.
9. Confirm live readback matches.
10. Test **Revert Last Remote Change**.

A good first test is a small change such as Wheel Damper `8 → 9`.

21. Appearance

Open **Customize Appearance** to change Atomic's visual theme.

Available theme presets include:

- Atomic Cyan
- Drift Orange
- Neon Purple
- Race Red
- Ice Blue
- Monochrome

Appearance changes are separate from tuning, telemetry, calibration, AZOM, and AC setup data.

22. Diagnostics

Open **System Diagnostics** if something is not working.

Diagnostics can help check:

- configured paths
- Assetto Corsa shared memory
- SimHub
- Atomic bridge
- AZOM visibility

- live base connection
- Atomic logs

Atomic can also export a privacy-redacted support ZIP for troubleshooting.

23. Recommended everyday workflow

1. Start SimHub + AZOM.
2. Start Atomic Drift Tuner.
3. Start Assetto Corsa.
4. Enter an on-track session.
5. Let Atomic auto-detect the car and pack.
6. Confirm wheelbase + steering wheel.
7. Choose Drift Target.
8. Generate Tune.
9. Review the recommendations.
10. Apply AZOM settings only if you want them.
11. Open AC Car Setup Tuner if you want setup changes.
12. Drive.
13. Record telemetry if needed.
14. Use Tuning Assistant + feedback to refine.
15. Save Desired Behavior for cars that need a specific feel.
16. Use Atomic Remote when you want phone access while in the rig.

24. Recommended first-time testing workflow

1. Let Atomic detect the car and pack.
2. Use **Neutral** Desired Behavior.
3. Choose your Drift Target.
4. Generate Tune.
5. Review AZOM + AC recommendations.
6. Apply one group of changes at a time.
7. Drive several laps.
8. Record telemetry.
9. Note what you want changed.
10. Adjust Desired Behavior or use Tuning Assistant.
11. Generate again.
12. Compare.
13. Save the setup once it feels right.

Avoid changing every possible variable at once. Smaller controlled changes make it easier to understand what helped.

25. Troubleshooting

Atomic does not detect the active car

Check that Assetto Corsa is actually on track, the AC path is correct, auto-scan is enabled, auto-select active AC car is enabled, and the car exists under the configured AC `content\cars` folder.

If the car is detected but the pack is unknown, **Custom / Other** may be correct.

Windows telemetry is offline

Make sure Assetto Corsa is running and you are in an on-track session.

Phone cannot load Atomic Remote

Check that the PC and phone are on the same LAN, use the LAN IP shown in Atomic instead of `127.0.0.1`, allow Atomic through Windows Firewall on the private network, and make sure router/AP client isolation is not blocking local devices.

Phone loads but telemetry is offline

Check the Windows **Shared AC telemetry** status inside the Remote window.

If Windows shows LIVE, the issue is likely in the browser/API path. If Windows shows OFFLINE, fix the AC telemetry connection first.

Live AZOM does not work

Check that SimHub is running, AZOM is enabled, the MOZA base is connected, Atomic SimHub Bridge v0.7.2 is installed and enabled, and Atomic can reach the local bridge.

AZOM apply fails verification

Atomic stops the batch when a setting cannot be verified.

Read live AZOM again, inspect the failed setting, and test it individually rather than repeatedly retrying a large batch.

26. Safety

Direct-drive wheelbases can produce significant force.

Always review recommendations before applying, make small changes first, stop testing immediately if the wheel behaves unexpectedly, and use Revert when needed.

Atomic is a tuning assistant. The final responsibility for the setup and safe operation of the hardware remains with the user.

27. Current beta notes

Current app version:

Atomic Drift Tuner v0.8.0-beta.1

Required Atomic SimHub Bridge for live AZOM features:

v0.7.2

Atomic Remote is currently intended for:

trusted same-LAN/private-network use

It should not be exposed directly to the public Internet.

Quick Start

If you just want the shortest possible workflow:

1. Open SimHub + AZOM.
2. Open Atomic.
3. Open Assetto Corsa and get on track.
4. Let Atomic auto-detect the car and pack.
5. Confirm wheelbase + wheel.
6. Choose Drift Target.
7. Click **Generate Tune**.
8. Review recommendations.
9. Apply AZOM changes only after review.
10. Use **AC Car Setup Tuner** for in-game setup changes.
11. Drive and record telemetry.
12. Use **Tuning Assistant** to refine.
13. Use **Atomic Remote** if you want live phone access.